Learning, Understanding & Modeling for Emerging Materials for Space Photovoltaics (LUMENS-PV)

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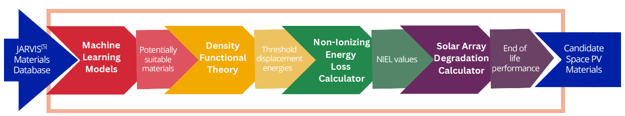
DSI DATA SCIENCE & INFORMATICS

Introduction

The integration of renewable energy sources, especially **photovoltaics** (PV), is important for **space applications** like satellites and space stations. Current PV materials are not well suited to last long due to the harsh conditions of space, which include extreme temperature fluctuations, high solar radiation levels, and a different solar spectrum.

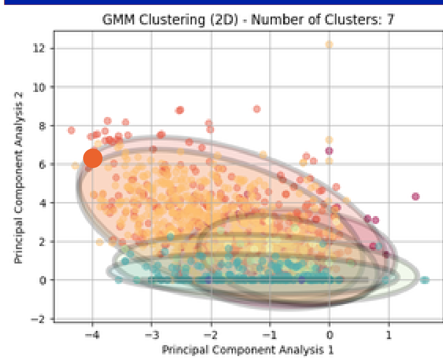
Our goal is to **accelerate the discovery of materials** using a combination of **machine learning** and **quantum-mechanical** methods to identify the best candidate materials for space-based PVs.



Our data comes from the **Joint Automated Repository for Various Integrated Simulations** (JARVIS) database that lists about 9,770 materials that have **Spectroscopic Limited Maximum Efficiency** (SLME) values > 10%. SLME has been used to classify PV materials for Earth's atmosphere [1].

Unsupervised Machine Learning Models

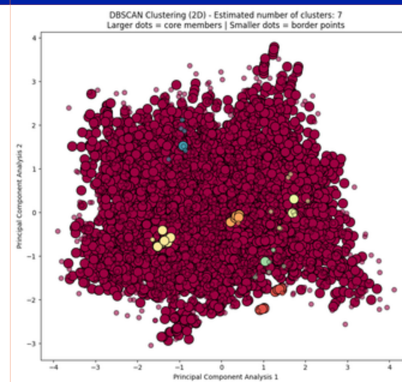
Gaussian Mixture Model (GMM)



GMM represents multi-variate data with distinct Gaussian distributions. They are especially ideal for datasets with unusual, non-uniform clusters. For our project, we employed GMM to determine if implicit variations resided within the data distributon.

We chose seven clusters based on **Bayesian Information Criterion** and optimization techniques. The **Silhouette Score** measures how well the data point fits in its assigned cluster, and our GMM model scored 0.041 with full covariance structure over 10 unique iterations.

Density-Based Spatial Clustering of Applications with Noise (DBSCAN)

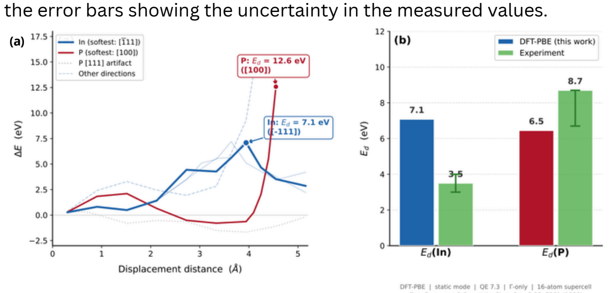


DBSCAN is a density-based clustering algorithm that is designed to discover clusters and noise in data. It identifies core, border and noise points. We used DBSCAN because there may be irregularities in our data that can create arbitrary-shaped clusters and it can identify and handle noise points[3]. Our DBSCAN model has a **Silhouette Score** of -0.117 with radius of 0.8 and min sample of 10.

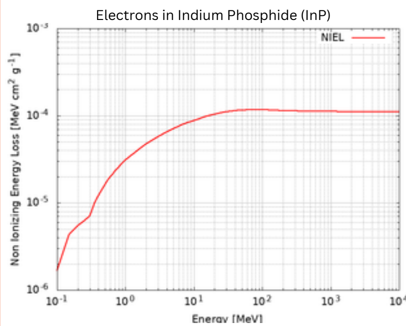
Density Functional Theory (DFT)

We use **DFT-PBE**[7] to calculate **threshold displacement energies** (E_d), the minimum energy needed to permanently displace an atom from its lattice site.

In panel (a), the plotted quantity is the change in total energy, ΔE, as the atom is moved away from its original position by a given displacement distance, measured in angstroms. These directional energy curves are then used to extract E_d. Panel (b) summarizes the final E_d values for In and P and compares them with experiment, with the error bars showing the uncertainty in the measured values.



Non-Ionizing Energy Loss (NIEL)



The red line represents the **NIEL values**[2], which is the amount of energy lost by a particle passing through a material that was imparted to create atomic displacements. As energy increases, it quantifies the displacement damage in the material.

The NIEL values are then passed to the OMERE Software[6] to analyze the impact of space radiation on the materials and electronic components in orbital environments. The **Solar Array Degradation Calculator** (SADC) is used and the active layer is replaced with the specific custom proton and electron NIEL values for the relevant material. The **Remaining Frequency** values are taken to represent the End of Life (EOL) values of the specific materials.

Future Directions

Future work will incorporate the **Spectroscopic Limited Practical Efficiency (SLPE) metric**[8] to more precisely quantify radiation-induced degradation in materials, particularly for space photovoltaics. Additionally, the study will improve ML and DFT models and expand to evaluate a larger and more diverse set of materials, enabling more comprehensive screening and identification of candidates with improved radiation tolerance for space applications.

References

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